

A gentle introduction to Bayesian statistics, with applications to bilingualism research

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Abstract

Bayesian analyses have been increasingly adopted in psychology and linguistics as an addition (or replacement) to traditional frequentist methods. However, Bayesian methods are not yet widely applied in bilingualism research, possibly because existing introductions and tutorials have not been directed specifically at our field. The current paper highlights the advantages of Bayesian statistics to the bilingualism researcher, by providing both an introduction to its foundational principles and a practical tutorial on estimation and hypothesis testing using the *brms* R package. The examples build up from simple linear regression to more advanced mixed-effects models and showcase different aspects of a Bayesian workflow. All data, code, models, and supplementary materials are publicly available at <https://osf.io/n3jgm/>.

Keywords: Bayesian statistics; bilingualism; Bayes factors; *brms*; tutorial

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Introduction

The last decade has seen some dramatic methodological changes in psychology and linguistics, primarily as a response to concerns that many experimental results may not generalise to participants, items, and tasks beyond those employed in particular studies (e.g., Gibson & Fedorenko, 2013; Maxwell et al., 2015). One area where such changes have been extensive concerns the statistical tools that we use to draw inferences about experimental data. For example, by-subject and by-item analyses have now been broadly replaced by mixed-effects models (Baayen et al., 2008). Another recent development is the increasing adoption of Bayesian statistics as an addition, or even replacement to traditional (frequentist) methods. Indeed, Bayesian methods now make regular appearances in psychology and linguistics journals (Van De Schoot et al., 2017), especially since recent advances in computational hardware and software have made it possible to fit Bayesian models more quickly and easily (Bürkner, 2017; Vasishth et al., 2018).

Nevertheless, and despite the fact that other methodological changes have already trickled down to bilingualism and second language research (Linck & Cunnings, 2015; Plonsky et al., 2021; Veríssimo, 2021b), Bayesian methods are not yet widely applied in our field. One reason for this slow adoption may be that Bayesian tutorials have predominantly targeted the psychology and psycholinguistics communities (e.g., Kruschke, 2015; van de Schoot et al., 2014; Vasishth et al., 2018) and thus may have escaped the notice of researchers in bilingualism (but see Garcia, 2024; Norouzian et al., 2018). A second reason is that getting started with Bayesian statistics can be challenging, because it is an altogether different framework from the one most researchers have been trained on. Indeed, many of the concepts that underlie our statistical practice, such as significance testing, p values, and Type I errors, are incompatible with a Bayesian approach. As a consequence, Bayesian methods are often perceived as having a steep learning curve—one that might not be worth climbing unless its benefits are made clear.

The current paper aims to fill these gaps. Instead of labouring over the long-standing debates between Bayesianism and frequentism within statistical theory (Bayarri & Berger, 2004; Mayo, 2018), we will take a more pragmatic stance. Our goal is not to ‘convert’ readers to Bayesianism (there are certainly many reasonable applications of frequentist statistics), but rather to provide a gentle introduction to the principles and practice of Bayesian statistics, while highlighting its potential uses for the bilingualism researcher.

The theoretical introduction is organised around three fundamental principles of Bayesian statistics: (a) model parameters are associated with full probability distributions, called posterior distributions; (b) these distributions express the probabilities of parameter values given the data, and (c) posteriors are obtained by combining prior knowledge with the evidence provided by the data. These principles will be demonstrated through the analysis of real datasets, building up from simple linear regression to more advanced mixed-effects models, while showcasing the different steps of a Bayesian analysis workflow. Throughout,

Table 1
Dataset example (from Veríssimo et al., 2018)

Subject	Inf.Priming	AoA.Ger
501	-105.82	5
502	104.01	5
503	-17.62	6
504	30.50	5
505	146.58	5
507	28.80	5

we will make use of the the *R* programming language and *brms* (version 2.20.4, Bürkner, 2017), an easy-to-use package for fitting Bayesian regression models.

Fundamental Bayesian principles

Principle 1: Parameters are associated with full probability distributions

The principles of Bayesian statistics are best understood in the context of a real dataset. The data for this example comes from a study with L2 learners of German in which morphological priming was predicted by age of L2 acquisition (AoA) (Veríssimo et al., 2018). Participants were presented with 50 ms subliminal primes that were inflectionally related (e.g., *gewandert* ‘hiked’) or unrelated (e.g., *komplett* ‘complete’) to target words (e.g., *WANDERN* ‘to hike’), and they performed lexical decisions on the targets.

We first read in the data:¹

```
head(data.priming <- readRDS("data/verissimo_2018_aoa5.rds"))
```

The first few rows of the data are shown in Table 1. The data has been aggregated, with each row in the dataset corresponding to one single participant. The “Inf.Priming” column shows the magnitude of inflectional priming for each participant, which was calculated by subtracting the participant’s mean response time (RT) on trials in the inflectional condition from their mean RT on the unrelated trials (the difference was calculated on logged RTs and then back-transformed to milliseconds to normalize the distribution). The “AoA.Ger” column shows each participant’s AoA of German, in years.

Note that, following current standards in psycholinguistics, this dataset would commonly be analysed with mixed-effects regression, thus obviating the need for aggregation. Moreover, the effect of AoA on priming would typically be assessed as an interaction between two variables (AoA $\times$ prime type), whereas in the current dataset, by-participant priming effects have already been calculated and serve as the dependent variable of interest. These

¹ Here, we analyse a subset of 47 subjects with an AoA greater than 5. The original paper reported a more complex non-linear pattern on a larger dataset (Veríssimo et al., 2018).

Table 2
Summary of frequentist model

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.45	10.52	0.61	0.543
c10(AoA.Ger)	-36.46	14.19	-2.57	0.014

modifications were made for reasons of simplicity, since they allow us to introduce the principles of Bayesian statistics with a simple one-predictor regression, before proceeding to more advanced mixed-effects regressions later in the paper.

We will first perform a frequentist regression on this data. For interpretability, AoA is divided by 10 and centred:

```
c10 <- function(x) scale(x/10, scale=F)      # Centring function
m.priming.freq <- lm(Inf.Priming ~ c10(AoA.Ger), # Frequentist regression
                    data=data.priming)
summary(m.priming.freq)                     # Model summary
```

Table 2 displays the summary of the frequentist model. It shows two coefficients: (a) a non-significant intercept that corresponds to the mean inflectional priming effect in this sample of L2 speakers, and (b) a significant effect of AoA, with inflectional priming decreasing as AoA increases, at a rate of about 36 ms per decade. Additionally, the model estimates a third parameter, *sigma*, which is the standard deviation (SD) of the residuals:

```
sigma(m.priming.freq)
```

```
## [1] 72.1
```

Together, the model's parameters (intercept, slope, and sigma) determine a normal distribution whose mean changes with the values of AoA. In a frequentist model, each of these parameters is estimated as a point value (the 'maximum likelihood estimate') accompanied by a measure of its uncertainty (the SE), from which a test statistic (the *t*-value) and a *p*-value can be calculated.

We now fit a Bayesian linear regression model. This can be easily achieved with the *brms* package, because its `brm()` function takes similar formulas to those used in frequentist models:²

```
m.priming <- brm(Inf.Priming ~ c10(AoA.Ger), # Bayesian regression
                data=data.priming)
summary(m.priming)                         # Model summary
```

Table 3 displays the summary of the Bayesian model, which shows the same three parameters (intercept, slope, and sigma). There are some notable differences, such as the

² Although fitting Bayesian models is typically straightforward, installing the *brms* package may require multiple configuration steps, depending on the user's system. Detailed instructions can be found in Coretta et al. (2023).

Table 3
Summary of Bayesian model

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	6.47	10.36	-14.34	26.59
c10AoA.Ger	-36.26	14.51	-64.63	-7.57
sigma	72.76	7.72	59.45	89.96

absence of t - and p -values, but the estimates and SEs are very similar to those obtained in the frequentist model. We can visualise the decrease in inflectional priming over AoA by plotting the Bayesian model's predicted regression line (Fig. 1):

```
ggemmeans(m.priming, terms="AoA.Ger") |> # Predictions
plot(show_data=T)                        # Plot
```

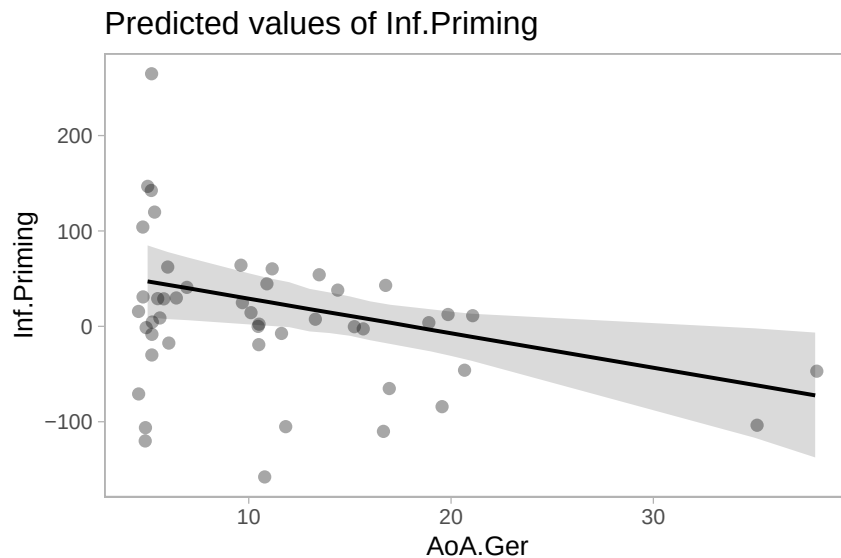


Figure 1. AoA effect on inflectional priming, as predicted by a Bayesian model (with flat priors).

However, the model summary and regression line obscure an important difference between Bayesian and frequentist approaches: in Bayesian models, estimates are not single fixed values, but full probability distributions. Specifically, each parameter is associated with a *posterior distribution* that captures the probability of different parameter values. This distribution is made up of samples collected during the fitting process (for more on sampling and assessing convergence, see Appendix S1). We can extract these ‘posterior samples’—by default, 4,000 for each parameter—and visualise them, as such:

```
ps.priming <- as_draws_df(m.priming)                # Get posterior samples

# Select parameters (fixed effects are prefixed by "b_")
pivot_longer(ps.priming, cols=c(starts_with("b_"), "sigma")) |>
# Plot
```

```
ggplot(aes(value)) + geom_density() +  
facet_wrap(vars(name), scales="free")
```

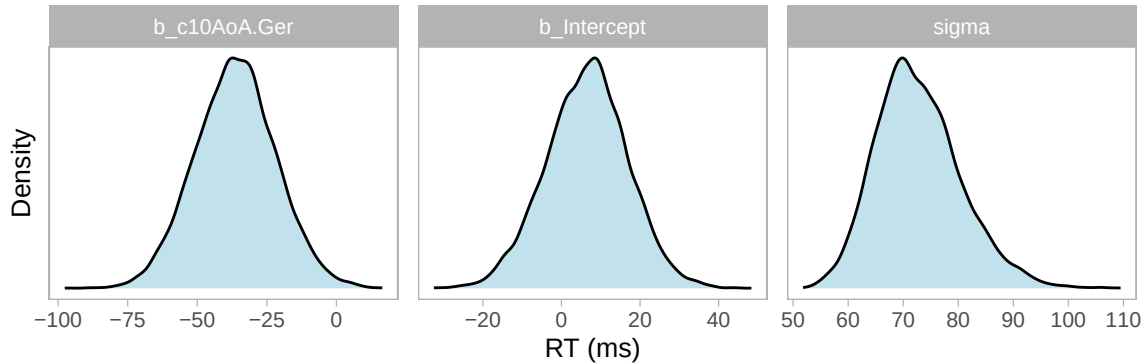


Figure 2. Posterior distributions for AoA slope (left), intercept (middle), and SD of residuals (right).

The resulting plots are displayed in Figure 2. The leftmost panel shows that the most probable values for the AoA slope are between -30 and -40 ms, but many other values are still probable. Thus, a statement such as “the regression slope of the AoA effect is -36 ms”—which would be appropriate in the case of a frequentist model—expresses only the mean of the posterior distribution, but falls short of capturing the full range of values estimated by the Bayesian model. Likewise, the regression line in Figure 1 is only one (the ‘most probable’) of multiple predicted regression lines.

Principle 2: Posteriors are probabilities of parameter values given the data

One of the main differentiating aspects of Bayesian statistics is that one can inspect the posterior distributions in Figure 2 to answer questions such as “what is the probability that the decrease in inflectional priming is between -40 and -30 ms per decade of AoA?” or “which are the most probable values of the AoA effect, with 95% probability?”. Frequentist approaches, in contrast, cannot provide such probabilities.

The frequentist p -value, in particular, is commonly misinterpreted as the probability of the null hypothesis (e.g., a p -value of .05 is interpreted as 5% probability that the null hypothesis is true). This is incorrect, since p -values are not probabilities of particular hypotheses or parameter values. Rather, a p -value is the probability of obtaining the data (strictly, of obtaining a test statistic that is as extreme or more) given that the null hypothesis is true (e.g., Wagenmakers et al., 2008). These are two very different questions: the one that the p -value answers goes something like “How impressed am I by this data, assuming there is no effect?”, $P(\text{data} | H_0)$, whereas the one we might want to see answered goes “How likely is it that there is no effect, given that I have obtained this data?” (i.e., $P(H_0 | \text{data})$). Humans—even seasoned scientists—tend to confuse the two, but they are demonstrably very different: the probability of being a UK citizen given that one is the UK prime minister is 1, but the probability of being the UK prime minister given that one is a UK citizen is close

to 0. The limitation is not specific to p -values. Confidence intervals (CIs) are also commonly misinterpreted as reflecting the range within which a true parameter lies, whereas the correct interpretation is that, under hypothetical replication, 95% of all such intervals are expected to contain the true value (e.g., Morey et al., 2016). Again, our desire to obtain probabilities of parameter values leads to a fallacy in which a conditional probability is mistaken for its inverse.

In contrast, the analogous Bayesian intervals can indeed be interpreted as the bounds within which a parameter lies. These are called ‘credible intervals’ (CrIs) and can be easily obtained as the quantiles of the posterior distribution. For example, to obtain the bounds of the 95% CrI for the AoA slope, we run:

```
ps.priming$b_c10AoA.Ger |>
  quantile(c(0.025, 0.975))
```

```
##    2.5%  97.5%
## -64.63  -7.57
```

We can interpret the 95% CrI as indicating that there is 95% probability that the AoA slope lies between -65 and -8 ms per decade. Note that the width of the 95% CrI—in fact, the width of the posterior distribution—naturally expresses our *uncertainty* around the parameter of interest. If the posterior is very wide, then many different values are possible, which is to say that our estimate is very uncertain. If the posterior is narrow, it means the parameter can be estimated with greater precision, since we are confident that its value lies within a smaller range of magnitudes.

Principle 3: Belief updating through the integration of prior knowledge

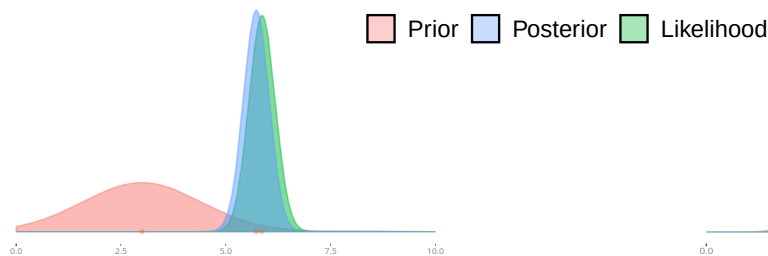
How can Bayesian credible intervals express the probabilities of parameter values if frequentist confidence intervals lack this property? The procedure that allows it, now known as ‘Bayes’ theorem’, consists of a rule that produces a conditional probability from its inverse:

$$P(\theta | y) \propto P(y | \theta) \times P(\theta)$$

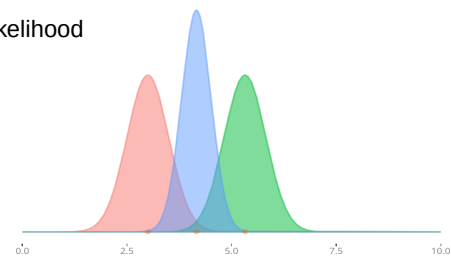
To see how Bayes’ theorem is applied, let us go back to estimating how inflectional priming changes with AoA. The AoA regression slope is our population parameter θ . In order to estimate θ , we collect some data y (in this case, priming effects for each participant). Bayes’ theorem yields the probability of different parameter values given the data, $P(\theta | y)$; this is the posterior distribution of the AoA slope (as shown in Figure 2). To obtain this posterior, two ingredients were needed. First, the probability of the data given each parameter value, $P(y | \theta)$; this is the familiar ‘likelihood’ of frequentist methods. Second, a distribution of parameter values, $P(\theta)$, which is defined independently the data; this is the *prior distribution*. The prior expresses our beliefs before seeing the data, the resulting posterior expresses our beliefs after considering the data, and Bayes theorem is the rule that formalises how our beliefs were updated from the evidence.

Figure 3 shows four different examples of how different priors and likelihoods combine to yield posteriors (for simplicity, normal distributions are assumed throughout). Panel A shows a situation in which a weaker (wider) prior is combined with a stronger (narrower) likelihood. Because there is more information coming from the data than from prior, the posterior is essentially determined by the data (albeit nudged in the direction of the prior). This is a common situation in Bayesian modelling, because we often use priors that are not very informative, but only slightly constraining. In Panel B, prior and likelihood have similar strength (i.e., they are equally narrow), so the posterior reflects the contribution of both. This would be a situation in which our prior beliefs are quite informative, for example, due to there being evidence from previous studies or meta-analyses indicating that certain parameter values are much more likely than others. Panel C depicts a similar situation, but the likelihood is now further away from the prior (i.e., takes higher values). This would be a situation in which the observed data is particularly surprising, since we *a priori* believed the observed effect to be very unlikely; however, because the data is strong enough (as strong as the prior), the posterior still reflects a compromise between them. In Panel D, we have a much weaker likelihood, but maintain a strong prior. This would be a situation in which the data provides very little information relative to our prior beliefs; perhaps the data is noisy or comes from a very small sample, so our beliefs remain virtually unchanged (i.e., the posterior is almost identical to the prior).

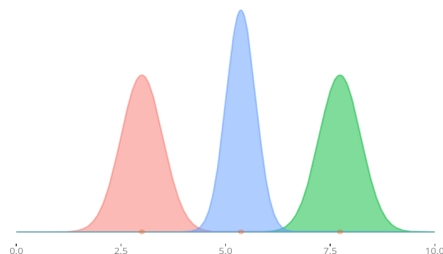
A. Weak prior, strong likelihood



B. Strong prior, strong likelihood



C. Strong prior, strong likelihood (larger mean)



D. Strong prior, weak likelihood

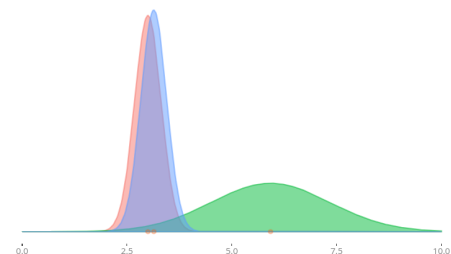


Figure 3. Four situations in which prior distributions and likelihoods combine to yield posterior distributions. See <https://mcl.shinyapps.io/prior2post/> (Clark, 2022) for an interactive version.

Let us go back to the Bayesian regression model that was previously fitted. We have obtained posteriors, so per Bayes' theorem, priors must have been employed. Prior distributions can be extracted from a fitted model as such:

```
prior_summary(m.priming) |> print(show_df=FALSE)

## b ~ (flat)
## b_c10AoA.Ger ~ (flat)
## Intercept ~ student_t(3, 7.4, 49)
## <lower=0> sigma ~ student_t(3, 0, 49)
```

The output shows that *brms* employed a 'flat' prior on the AoA slope (inherited from a general prior on slopes, "b"). Flat priors are uniform distributions in which every value in the real line is assumed to be as likely as any other before seeing the data. Such priors should not generally be used in Bayesian modelling, for various reasons: (a) they make very unrealistic assumptions about possible effects (e.g., that a decrease in morphological priming of a few milliseconds per decade of AoA is as likely as a decrease of 1 second, 1 hour, or 1 day); (b) they can lead to posterior inferences that are particularly extreme, for instance, due to outliers in the data; (c) they make model convergence more difficult, especially for complex models; (d) they make it impossible to perform hypothesis testing via Bayes factors (see below). In addition to the prior on slopes, *brms* has also automatically assigned student-*t* priors on intercept and sigma, on the basis of the variance of the data. Since priors capture our beliefs before observing the data, it is also best to avoid such automatically-defined priors.

One way to improve this model is by setting priors on all three parameters. Each parameter belongs to a different prior 'class', "Intercept" for the regression intercept, "b" for slopes, and "sigma" for the residual SD:

```
priors.priming.weaker <-
  prior(normal(0, 250), class=Intercept) +
  prior(normal(0, 250), class=b) +
  prior(normal(0, 250), class=sigma)
```

We set the same prior on all three parameters, a normal distribution with mean 0 and SD 250.³ Since a normal distribution has 95% of its mass between the mean plus or minus 2 SDs, a Normal(0, 250) prior expresses that both intercept and slope are between -500 ms and 500 ms with 95% probability. Since masked priming effects are not generally larger than the prime duration (in this case, 50 ms) (Forster et al., 2003), these priors can be considered only weakly informative. We refit the model with the new priors:

```
m.priming.weaker <- brm(Inf.Priming ~ c10(AoA.Ger),
  data=data.priming,
  prior=priors.priming.weaker)
```

³ The prior on sigma is actually turned into a 'half-normal' distribution with a lower bound at 0, because an SD cannot be negative.

Next, we examine the estimated fixed effects:

```
fixef(m.priming.weaker)
```

```
##           Estimate Est.Error  Q2.5 Q97.5
## Intercept      6.43      10.9 -14.8 28.12
## c10AoA.Ger    -36.33     14.7 -65.2 -7.18
```

The estimates are very similar to those obtained with flat priors (cf. Table 3). This is because with weakly informative priors, the posteriors are still almost completely determined by the likelihood (a situation similar to that in Fig. 3A).

Can our model be improved by setting more informative priors? Indeed, one of the most beneficial aspects of Bayesian statistics is that better inferences can generally be obtained by taking prior information into account. To demonstrate this, let us examine the predicted AoA effect in Figure 1 (in this case, from the model with flat priors). It is noticeable that the regression line extends into a large inhibitory effect at the higher AoAs (-74 ms at an AoA of 38), a prediction that seems implausibly large when we consider typical magnitudes of masked priming effects in L1 and L2 (e.g., Feldman et al., 2010; Silva & Clahsen, 2008). Moreover, there are only two participants with an AoA over 30, so the regression line may have been unduly influenced by them.

We can constrain the model's predictions by setting more realistic (i.e., narrower) priors on intercept and slope, which requires formalising our expectations about priming magnitudes. We start by examining masked priming effects more generally (i.e., not only morphological) and find that effects, whether facilitatory or inhibitory, are not generally larger than the prime duration (Davis & Lupker, 2006; Forster et al., 2003). We then consult meta-analyses of masked morphological priming in L1 (Rastle & Davis, 2008) and L2 (Fernandes et al., 2023), which show that derived words (not inflected forms, as in the current analysis) show mean effects ranging from 18 ms to 44 ms in L1 and -4 ms to 60 ms in L2. Finally, we examine previous studies on masked *inflectional* priming in L2 and find that these have yielded heterogeneous results, including both inhibitory and facilitatory effects of different magnitudes (from -27 ms to 52 ms, e.g., Feldman et al., 2010; Foote, 2017; Pliatsikas et al., 2014; Silva & Clahsen, 2008).

On the basis of this literature review, we assign a Normal(0, 25) prior on the regression intercept, expressing that it is very likely (with 95% probability) that the mean inflectional priming effect in L2 learners is between -50 ms and 50 ms. Likewise, we assign a Normal(0, 25) prior on the AoA slope, which places 95% probability on the AoA effect being between -50 ms and 50 ms per decade. This prior would be consistent with large qualitative differences between early and late bilinguals, for example, with a facilitatory effect of 50 ms at an AoA of 5, no effect at an AoA of 15, and a -50 ms inhibitory effect at an AoA of 25. Thus, even though these priors are much more informative, they still admit a wide range of magnitudes and do not completely rule out very large effects:

```
priors.priming.stronger <-
  prior(normal(0, 25), class=Intercept) +
  prior(normal(0, 25), class=b) +
  prior(normal(0, 250), class=sigma)
```

We fit a model with the stronger set of priors:

```
m.priming.stronger <- brm(Inf.Priming ~ c10(AoA.Ger),
  data=data.priming,
  prior=priors.priming.stronger)
```

The posterior means of intercept and slope are smaller than in the model with weaker priors:

```
fixef(m.priming.stronger)
```

```
##           Estimate Est.Error  Q2.5 Q97.5
## Intercept      5.41      9.95 -14.3  25.11
## c10AoA.Ger    -27.11     12.73 -51.7  -1.66
```

We can plot the two regression lines simultaneously, as such:

```
# Predictions (weaker priors)
line.weaker <- ggemmeans(m.priming.weaker, terms="AoA.Ger")

# Predictions (stronger priors)
ggemmeans(m.priming.stronger, terms="AoA.Ger") |>
  plot() + geom_line(data=line.weaker,
    aes(y=predicted, x=x),
    col="red")
```

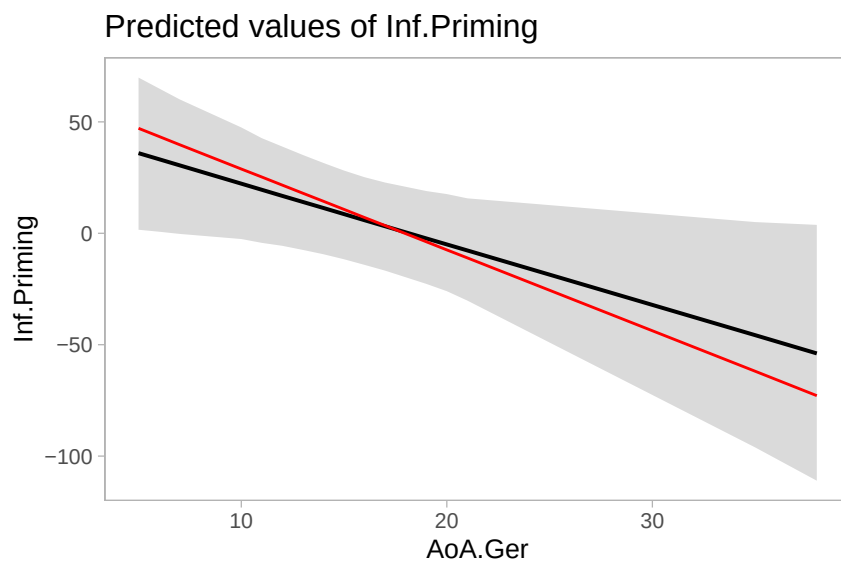


Figure 4. AoA effect on inflectional priming, as predicted by Bayesian models with weaker (red) and stronger (black) prior distributions.

The predictions are similar in the two models (Fig. 4, but in the model with stronger priors (black line), the regression line is pulled towards zero. Our predictions at the higher AoAs are now more constrained (-54 ms at an AoA of 38) and less influenced by extreme data points. Thus, even though this is no longer the ‘best-fitting line’ (in the maximum-likelihood sense), it is arguably a better and more generalisable prediction.

Bringing priors into the open. Newcomers to Bayesian statistics often feel uneasy about the use of priors in statistical inference, for various reasons. First among these is that defining priors is seen as a difficult and uncertain task—one that requires a certain amount of (educated) guessing about plausible effects, which is a skill that most researchers have not been trained on. Fortunately, the practice of prior selection can be made much less daunting by incorporating a few principles and guidelines into the Bayesian analysis workflow (Nicenboim et al., 2024; Schad et al., 2021): (a) weakly-informative priors can be constructed by employing wide distributions that rule out only the most unreasonable and extreme values; (b) to define more informative priors, one can consult available meta-analyses and systematic reviews to obtain means and ranges of relevant effects (as was done above); even when the previous literature does not test exactly the same manipulation or group of speakers, the reported effects can still be used as a general guideline; (c) priors can also be defined by asking oneself or others how surprising certain effects would be, and then placing the most unexpected results at the tail of the prior distribution (for a systematic approach, see O’Hagan, 2019); and finally, (d) if a model contains many parameters, the use of prior predictive checks (demonstrated below) is an essential tool for examining the plausibility of the data predicted by a given set of priors.

Another concern about the use of priors is that they make the analytic process less impartial or ‘scientific’, because a researcher’s subjective beliefs are allowed to alter the outcome of the analysis. We would argue that this is actually a good thing, since the goal of a statistical analysis is not to characterise what happens in any given sample, but to arrive at the best possible inference. Incorporating prior information gets us closer to that goal, because—as shown above (Fig. 4)—when a dataset includes particularly extreme values, a model with weaker priors (or a frequentist model) will tend to overfit the data, while a model with stronger priors can yield more plausible inferences (Gelman & Tuerlinckx, 2000; Wagenmakers et al., 2008).

As for the subjectivity of priors, we note that prior choice is no different from other decisions in statistical modelling, such as whether or not to remove outliers, transform a dependent variable, or include covariates in a regression model. And as is the case for those analytical decisions, priors should be transparently reported, reasonably justified, and capable of withstanding scrutiny. Moreover, researchers always bring prior knowledge into their interpretations, even in frequentist analyses (Nicenboim et al., 2024; Wagenmakers et al., 2008): outcomes that are obvious or highly expected are more believable—precisely because they are consistent with what is already known—whereas outcomes that are surprising

or that challenge the literature are less likely to be believed as general claims. The main difference is that, in frequentist statistics, prior beliefs are operating inside the minds of researchers; in Bayesian statistics, we are forced to bring our priors into the open and explicitly integrate them into the analysis.

Estimation and hypothesis testing

One consequence of the Bayesian principle that posterior distributions represent probabilities of parameter values is that this directs our attention to the *magnitudes* of experimental effects. Kruschke and Liddell (2018) describe two contemporary changes in statistical practice, a shift from frequentist to Bayesian statistics, and a shift from hypothesis testing to estimation. The two shifts do not completely overlap, since estimation can be performed in a frequentist framework and hypothesis testing can be conducted in a Bayesian framework. However, Bayesian methods align particularly well with the estimation goal, because posteriors express a complete probability distribution over the credible values of a parameter (Kruschke & Liddell, 2018; Vasishth et al., 2018). In contrast, hypothesis testing under the frequentist framework often devolves into a binary decision that merely classifies effects as significant or not (Cohen, 1994; Cumming, 2014).

From an estimation perspective, the AoA effect presented above can be described as having a mean credible value of -27 ms per decade, with a 95% probability that this reduction is between -52 ms and -2 ms. We could also summarise the posterior distribution with other measures of central tendency or other credible regions. For example, if we wanted to determine the posterior mode (the most probable value for the parameter) and the 90% highest-density intervals (the smallest contiguous region of credible values with 90% probability), we would load the *ggdist* package and run:

```
ps.priming.stronger <- as_draws_df(m.priming.stronger)
point_interval(ps.priming.stronger$b_c10AoA.Ger,
               .point=Mode, .interval=hdci, .width=.90)
```

```
##           y  ymin  ymax .width .point .interval
## 1 -27.5 -48.6 -6.75    0.9   mode      hdci
```

We can also calculate the proportion of the posterior that goes in the expected direction (the ‘probability of direction’, Makowski et al., 2019), or that is larger in absolute magnitude than a ‘smallest effect size of interest’ (SESOI, Lakens et al., 2018) (e.g., 5 ms/decade, if we considered a smaller effect to be scientifically irrelevant):

```
# Probability of effect being negative
sum(ps.priming.stronger$b_c10AoA.Ger < 0) / length(ps.priming.stronger$b_c10AoA.Ger)
```

```
## [1] 0.982
```

```
# Probability of effect being smaller than SESOI
sesoi <- -5
sum(ps.priming.stronger$b_c10AoA.Ger < sesoi) / length(ps.priming.stronger$b_c10AoA.Ger)
```

```
## [1] 0.957
```

Since the vast majority of credible values are negative (and large enough to matter), it seems quite likely that each year of AoA reduces inflectional priming by at least a couple of milliseconds (at the same time, it is still possible that the true effect goes in the opposite direction, albeit with only 1.9% probability). When examined against the broader context of research on masked morphological priming, the effect of AoA appears to be quite large. For example, if those who learned German at an age of 5 perform as L1 speakers—which, as we have seen, generally show priming effects of about 20-40 ms—then the priming effect would be expected to disappear completely for those who learned German at age 15.

Such estimation-based considerations are often more informative than testing an effect against a single null value. Had we instead asked if “there was an effect of AoA”, our inference would have been reduced to a single existence statement (i.e., AoA effect different from 0) and we would have ignored its magnitude and uncertainty. For these (and other) reasons, estimation has long been argued to be more in line with the goals of scientific inference than hypothesis testing (e.g., Cohen, 1994; Kruschke & Liddell, 2018; Nickerson, 2000).

Nevertheless, if testing an effect against a null value is deemed necessary to answer a research question (e.g., Morey et al., 2014), then it can also be performed with Bayesian statistics. Bayesian hypothesis testing is based on comparing an alternative model, which includes the effect of interest, to a null model, which does not. The comparison yields a quantity—called the ‘Bayes factor’—that quantifies the evidence that the data provides in favour of one versus the other hypothesis, and thus allows inferring about the existence of an effect (e.g., Dienes & Mclatchie, 2018; Rouder et al., 2009; Schmalz et al., 2023). In the case of point-null hypotheses (e.g., AoA slope equal to 0), the Bayes factor can be calculated using the alternative model only, since it is equal to the ratio of prior and posterior densities at the null value (Dickey & Lientz, 1970):⁴

```
(hyp.priming.stronger <- hypothesis(m.priming.stronger, "c10AoA.Ger = 0"))
```

```
##           Hypothesis Estimate Est.Error CI.Lower CI.Upper Evid.Ratio
## 1 (c10AoA.Ger) = 0      -27.1      12.7    -51.7     -1.66      0.215
```

The Bayes factor in favour of the null hypothesis, BF_{01} , is 0.22 (shown in the “Evid.Ratio” column). We can obtain the Bayes factor in favour of the alternative model, BF_{10} , by computing the inverse of BF_{01} ; additionally, the Bayes factor is sometimes natural-logged ($\ln BF_{10}$), so that positive and negative values indicate support for one versus the other hypothesis:

```
bf01 <- hyp.priming.stronger$hypothesis$Evid.Ratio
log(1/bf01)
```

```
## [1] 1.54
```

⁴ Calculating Bayes factors in this way requires sampling from the priors (with `sample_prior="yes"`) and increasing the number of posterior samples (see Appendix S1).

In general, values of $\ln BF_{10}$ greater than about 1 provide some support for the alternative model, while negative values smaller than -1 support the null model; values between -1 and 1 are essentially inconclusive (Kass & Raftery, 1995; see Lee & Wagenmakers, 2013, for a similar scale of evidence). Unlike null-hypothesis testing in a frequentist framework, Bayes factors *can* provide support for the absence of an effect, that is, for the equality of conditions or groups, which is an important advantage of Bayesian statistics (Dienes & McIlatchie, 2018; Rouder et al., 2009; but see Lakens et al., 2018).

One interpretation of the Bayes factor is as a measure of ‘belief updating’, specifically, of how much our prior belief that the effect was absent has changed by observing the data. This means that Bayes factors can be very sensitive to the specific priors employed in the alternative model. For example, if we instead calculate the AoA effect Bayes factor from the model with weaker priors, the resulting $\ln BF_{10}$ shows only very slight support for the alternative hypothesis ($=0.15$). Given that Bayes factors can yield different conclusions under different priors, it is considered good practice to examine the relevant hypotheses under a range of reasonable priors, both weakly and strongly informative (Schad et al., 2022; Sinharay & Stern, 2002). In the case of the AoA effect, the Bayes factor is not completely robust to these prior specifications—but it could still be argued that the stronger priors are a better expression of our prior beliefs than the weaker ones, thus suggesting a non-zero AoA effect.

A final note about Bayesian hypothesis testing: when we calculate a 95% credible interval, it is enticing to inspect whether the interval crosses zero and claim the existence of an effect when it does not. This approach is borrowed from frequentist statistics, where there is a perfect mapping between rejecting the null hypothesis (with $p < .05$) and the corresponding 95% confidence interval not crossing zero. However, this relationship does not hold in Bayesian statistics: it is perfectly possible for the null value to be outside of a 95% credible interval—and thus be considered “improbable” in an estimation-based approach—while the Bayes factor actually provides support for the null hypothesis (Rouder et al., 2018; Wagenmakers et al., 2022). The reason for this apparent paradox is that Bayes factors and credible intervals are conceptually distinct. Bayes factors reflect the evidential weight for one versus the other hypothesis (i.e., effect present versus absent), while credible intervals provide the range of credible values under the alternative hypothesis, that is, assuming that the effect is present (Marsman & Wagenmakers, 2017; Wagenmakers et al., 2022). Credible intervals are extremely useful for quantifying effects (i.e., for the purposes of parameter estimation), but since they effectively ignore the null model as a competing hypothesis, they should not be used to make categorical claims about whether or not effects “exist” (i.e., for the purposes of hypothesis testing) (Nicenboim et al., 2024; Rouder et al., 2018).

Table 4
Dataset example (from Broos et al., 2021)

Participant	Group	Item	Task	Accuracy	RT
2	L2	coat	Regular	1	968
2	L2	coat	Delayed	1	382
2	L2	doll	Regular	1	968
2	L2	doll	Delayed	1	480
2	L2	knife	Regular	1	712
2	L2	knife	Delayed	1	392

Bayesian mixed-effects models

The analysis above served to illustrate the main principles of Bayesian inference, but it is admittedly not very representative of the analyses that are commonly conducted in our field. For one, many of our experimental tasks consist of multiple participants responding to multiple items. Aggregation by participant, as was done above, leaves out systematic item variance and can lead to biases in statistical inference (Schad et al., 2024). Moreover, the models above did not specify a response distribution family, which means that residuals were assumed to be normally distributed (if unspecified, the `brm()` function employs normally-distributed residuals as the default). However, for many data types, this assumption would be unwarranted, including right-skewed distributions (e.g., naming times, eye fixation times) and categorical outcomes (e.g., accuracies, rating scales). Such measures are best analysed with *generalized mixed-effects models*, which obviate the need for prior aggregation and can accommodate different response distributions (Baayen et al., 2008; Linck & Cummings, 2015).

To demonstrate a Bayesian mixed-effects analysis, we will use a dataset consisting of picture naming times. Broos et al. (2021) tested whether slower picture naming in L2 than in L1 is caused by delays in earlier or later processes. To do that, they asked 40 English monolinguals and 43 L2 speakers to name 24 pictures as quickly as possible, either immediately after picture onset (regular task) or after a cue that appeared 1250 ms post-picture (delayed task).

The first few rows of this dataset are displayed in Table 4, where it can be seen that the same participants and items were tested in both tasks (Table 4) (only correct responses are analysed here):

```
head(data.naming <- readRDS("data/broos_et_al_2021.rds"))
```

Fitting a Bayesian generalized mixed-effects model is very similar to fitting a simple linear regression, but requires a few additional steps. The first is to select an appropriate response distribution. A reasonable starting point given the typical right-skewness of naming times is a *lognormal* distribution. This produces similar results to transforming RTs by logging, but the data and the model's predictions can remain in the untransformed millisecond scale.

Second, mixed-effects models can be applied to unaggregated data because they make use of random effects, which are additional parameters that capture the variation in intercepts and slopes across participants and items (Baayen et al., 2008; Linck & Cummings, 2015). Due to these additional parameters, mixed-effects models typically take longer to fit and demand greater computational resources (see Appendix S2 for tips on how to increase the computational efficiency of Bayesian models).

Model specification

We will first fit a model with default priors to help us define an appropriate model specification. In this model, RTs are predicted by Task (regular, delayed) and Group (L1, L2). One can parametrise such a model in different ways. If the model formula was written as `Task * Group`, we would directly estimate the interaction between the two predictors. Here, we instead specify the model in a way that estimates two specific contrasts of interest, namely, the difference between the L1 and L2 groups, separately for the regular and the delayed task. To do so, we estimate ‘nested’ effects of group under task by writing the formula as `Task / Group` (Schad et al., 2020) (as we will see below, the difference between the two group effects, i.e., the task $\times$ group interaction, can be easily estimated after fitting the model). This model has a ‘maximal’ random-effects structure, that is, it includes all random slopes allowed by the design (Barr et al., 2013):

```
# Factor levels
data.naming$Group <- factor(data.naming$Group, levels=c("L1", "L2"))
data.naming$Task <- factor(data.naming$Task, levels=c("Regular", "Delayed"))

# Formula (defined within bf())
formula.naming <- bf(RT ~ Task/Group +           # Fixed effects
                    (1+Task|Participant) +       # Random effects (by participant)
                    (1+Task/Group|Item))         # Random effects (by item)

# Fit model (default priors)
m.naming.flat <- brm(formula.naming,
                     data=data.naming,
                     family=lognormal())
```

The model’s fixed-effect estimates can be obtained as such:

```
summary(m.naming.flat)$fixed["Estimate"]
```

```
##              Estimate
## Intercept          6.5503
## TaskDelayed        -0.4599
## TaskRegular:GroupL2  0.2731
## TaskDelayed:GroupL2 -0.0173
```

The intercept corresponds to the mean RT of the L1 group in the regular naming task, expressed in log-milliseconds (because these were the first levels of each predictor and, by default, R codes two-level factors as 0 and 1). The TaskDelayed coefficient is the difference between mean RTs in the delayed task and the regular task, for the L1 group. The last two coefficients express the difference between the L2 and L1 groups, separately for the regular and delayed tasks. Our goal with this model is only to understand what the parameters mean, so we will refrain from interpreting the coefficients at this stage.

The estimates of participant random effects can be obtained as such:

```
# By-participant random effects (replace with "Item" for item random effects)
summary(m.naming.flat)$random$Participant["Estimate"]

##              Estimate
## sd(Intercept)      0.1380
## sd(TaskDelayed)    0.2251
## cor(Intercept,TaskDelayed) -0.0161
```

The “sd(Intercept)” estimate is the SD of random intercepts, which expresses the variation across participants in mean naming times, in the regular naming task; the “sd(TaskDelayed)” estimate is the SD of random slopes, which expresses variation across participants in the effect of task (delayed vs. regular); the “cor(Intercept,TaskDelayed)” estimate shows the correlation between random intercepts and slopes (i.e., the extent to which participants with slower naming times in the regular task show a larger or smaller difference between tasks). Random effects by item could be obtained in the same way, but these consist of a larger number of parameters, because group effects can also be estimated within items. Given that the model is parametrised in a way that directly estimates the hypotheses of interest, we can proceed to assigning priors to each of the parameters.

Prior choice and model checks

We start with a set of very weak priors (all of which expressed in log-ms):

```
priors.naming.weaker <- # Weakly-informative priors (lognormal)
prior(normal(0, 10), class=b, coef=Intercept) +
prior(normal(0, 1), class=b) +
prior(normal(0, 1), class=sd, group=Participant, coef=Intercept) +
prior(normal(0, 1), class=sd, group=Item, coef=Intercept) +
prior(normal(0, 1), class=sd, group=Participant) +
prior(normal(0, 1), class=sd, group=Item) +
prior(lkj(2), class=cor) +
prior(normal(0, 1), class=sigma)
```

This set of priors shows several new elements. First, the “Intercept” is now specified as a coefficient of class “b”, rather than as a separate class. This is because, for reasons of efficiency, priors of class “Intercept” in *brms* are not set on the actual intercept, but on an ‘internal’ intercept that corresponds to the *overall mean* of the response variable—regardless of what the model intercept actually represents. We can bypass this internal conversion by

writing the prior as above (together with a change in the model formula; see below). In this way, the prior will apply to the actual model intercept, which as we have seen, is an estimate of mean RTs for the L1 group in the regular task. A second novel aspect is that there is a class of “sd” priors for the SDs of random effects, with random intercepts and slopes distinguished by the “coef” argument and defined separately for each “group” (i.e., Participant or Item). Finally, there is a prior of class “cor”, which is assigned to the correlations between random effects.

These priors were chosen to be extremely weak. A Normal(0, 10) prior on the intercept places 95% of its mass between $\exp(-20)$ and $\exp(20)$, and thus it allows extremely small and extremely large RTs (ranging from practically zero to about 135 hours). Likewise, the Normal(0, 1) prior on the other parameters is compatible with very large experimental effects (between -2 and 2 log-ms with 95% probability), as well as with large variation between participants and items. The prior on random correlations is an *LKJ* distribution of ‘shape’ 2, a weakly-informative prior on correlations (for a visualisation, see Appendix S3).

When a model contains a large number of fixed and random parameters, and moreover, if they are expressed in a less natural scale such as log-ms, it becomes harder to understand how the different prior distributions interact to express our expectations. A very useful procedure in this context is to generate simulated data from a model that has only ‘seen’ the priors but has not been fitted to any observed data. To conduct such *prior predictive checks*, we first fit a model that samples only from the prior distributions:

```
formula.naming <- bf(RT ~ Task/Group +
  (1+Task|Participant) +
  (1+Task/Group|Item),
  center=FALSE) # We prevent centring ...
                # ... to suppress the 'internal intercept'

m.naming.weaker.prior <- brm(formula.naming,
  data=data.naming,
  family=lognormal(),
  prior=priors.naming.weaker,
  sample_prior="only") # Sample from prior
```

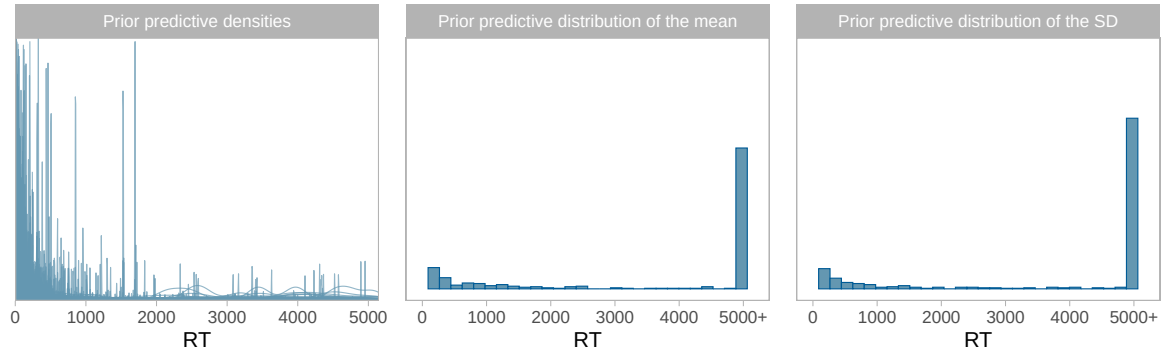
The prior predictive checks can then be conducted with `pp_check()`:

```
# Prior predictive distributions
pp_check(m.naming.weaker.prior, prefix="ppd") # Densities
pp_check(m.naming.weaker.prior, prefix="ppd", type="stat", stat="mean") # Means
pp_check(m.naming.weaker.prior, prefix="ppd", type="stat", stat="sd") # SDs
```

The results are shown in Figure 5A. The first facet shows overlaid density plots for each simulated dataset. They reveal that the model predicts impossible distributions, with many density peaks close to 0 ms and several other peaks scattered at different RTs. The second and third facets show histograms for the means and SDs of the simulated datasets.

The majority of means and SDs are above 5000 ms, which are implausibly large values in a naming task.

A. Lognormal, weaker priors



B. Lognormal, stronger priors

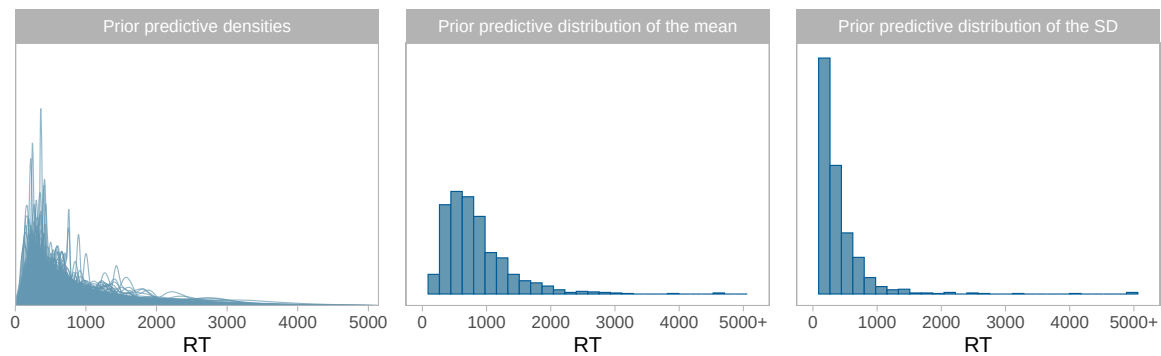


Figure 5. Prior predictive checks for lognormal models with (A) weaker priors, and (B) stronger priors. The prior predictive densities plots (left facets) have been clipped at 5000 ms to facilitate visualisation (but there are actually many simulated datapoints above that value). In the histograms of means and SDs (middle and right facets), the last bar includes all means/SDs greater than 5000 ms.

Since the weaker priors are clearly inadequate, we define a more informative set of priors that incorporates knowledge about plausible naming times and effect sizes. We start by consulting two different meta-analyses of picture naming (Bürki et al., 2020; Bylund et al., 2023). In Bürki et al.’s meta-analysis of picture-word semantic interference, the mean RTs of examined studies ranged from 613 ms to 1180 ms (mean: 765 ms); we may expect the mean RT in a straightforward picture naming task (without superimposed words) to be on the faster end of this range. In Bylund et al.’s meta-analysis of lexical access in bilingual production, mean picture naming times of monolinguals ranged from 482 ms to 1507 ms (mean: 808 ms). All considered, we opt for a Normal(6.5, 0.5) prior on the intercept, which is much narrower than the prior employed above, but still wide enough to capture a large range of naming times: it states that mean RTs (for the L1 group in the regular task) are likely to be between 5.5 to 7.5 log-ms (i.e., 245ms to 1808ms).

As for the priors on regression slopes, we again consult Bylund et al.'s (2023) meta-analysis to obtain the differences between mean RTs of late L2 learners and monolinguals in the examined studies. The differences range between -72 ms to 545 ms, with a mean of 111 ms. On this basis, we define a Normal(0, 0.25) prior on the fixed slopes, which is still wide enough to capture both small and large differences between the groups and tasks (to put it in perspective, an effect of 100 ms effect relative to a 700 ms baseline RT is 0.13 log-ms).

As above, we also need to assign priors to the random effect and residual SDs:

```
priors.naming.stronger <- # Stronger priors (lognormal)
prior(normal(6.5, 0.5), class=b, coef=Intercept) +
prior(normal(0, 0.25), class=b) +
prior(exponential(10), class=sd, group=Participant, coef=Intercept) +
prior(exponential(10), class=sd, group=Item, coef=Intercept) +
prior(exponential(10), class=sd, group=Participant) +
prior(exponential(10), class=sd, group=Item) +
prior(lkj(2), class=cor) +
prior(exponential(5), class=sigma)
```

A notable change is we have opted for exponential priors on the random effects and sigma. Exponential distributions are arguably more appropriate than normal distributions in this case, because they are positive-only distributions, and they place most of their mass at the smaller values, but decline gradually into the larger values (McElreath, 2020). Different exponential 'rates' create steeper or gentler declines, thereby encoding our expectations about different values. An exponential of rate 10 on random intercepts and slopes states that random SDs are unlikely to be much larger than about 0.4 or 0.5 log-ms (see Fig. 6). Random intercepts and slopes are themselves assumed to follow a normal distribution, so a by-participant random slope SD of 0.5 around a given effect would imply that 95% of participants have effects ranging from -1 to +1 log-ms relative to the average—indicating very large variation between participants. Finally, the prior on sigma has rate 5, because the residual SD is expected to be the largest source of variance (this is almost always the case in psycholinguistic experiments).

The prior predictive checks for the model with stronger priors are presented in Figure 5B. The density plots now take less extreme values and show the typical right-skewed shape of RT distributions in naming tasks. Moreover, the simulated datasets have much smaller (and more plausible) means and SDs. In sum, the model with stronger priors does a much better job at predicting the kind of data we might see in such an experiment. We fit the model on the observed data:

```
m.naming.stronger <- brm(formula.naming,
  data=data.naming,
  family=lognormal(),
  prior=priors.naming.stronger)
```

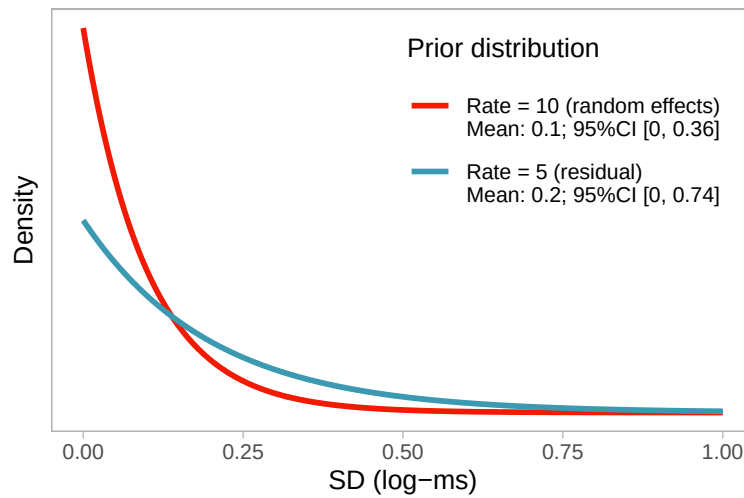


Figure 6. Exponential distributions used as priors for random and residual SDs (lognormal model, stronger priors).

Once a model is fitted, we should again run a set of predictive checks. However, these will now be *posterior* predictive checks, in which we simulate datasets from the model's posterior distributions and compare them against the observed data. If the simulated data does not align well with the observed data, this may indicate a violation of the model's assumptions. As above, predictive checks can be conducted for different statistics of interest:

```
# Posterior predictive distributions
pp_check(m.naming.weaker.prior)           # Densities
pp_check(m.naming.weaker.prior, type="stat", stat="mean") # Means
pp_check(m.naming.weaker.prior, type="stat", stat="sd")  # SDs
```

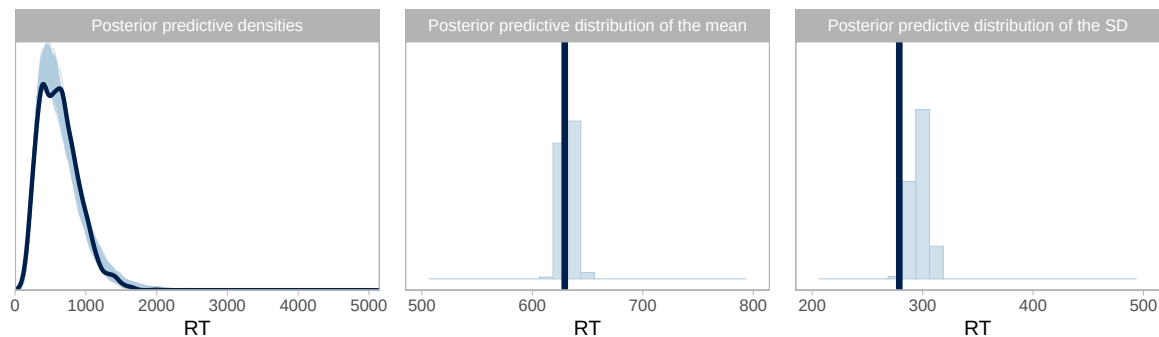


Figure 7. Posterior predictive checks (lognormal model, stronger priors).

The posterior predictive densities (Fig. 7, left facet) show that the model is able to capture the general shape of the distribution of responses. Likewise, the simulated means (middle facet) cluster around the observed mean, indicating a good fit to the data. Nevertheless, the fit is not perfect. The predictive densities show some discrepancies relative to the observed data, such as an overestimation of RTs at the peak density (i.e., around 300 ms–500 ms), and simulated SDs that are slightly larger than the observed SD (right facet). The

model could possibly be improved by incorporating additional explanatory predictors (e.g., language proficiency) and/or by fitting models that assume different response distributions (e.g., the shifted lognormal or the ex-Gaussian, as in Ciaccio & Verissimo, 2022; Matzke & Wagenmakers, 2009). Models belonging to different families can then be assessed by inspecting their posterior predictive checks, or formally compared (e.g., with cross-validation, Vehtari et al., 2017; Verissimo, 2021a).

Model predictions and inference from posteriors

If we are satisfied with the predictive checks, we proceed to examine the model's predictions (in milliseconds) in the different groups and tasks (Fig. 8):

```
ggemmeans(m.naming.stronger, terms=c("Task", "Group"), epred=T) |>
  plot()
```

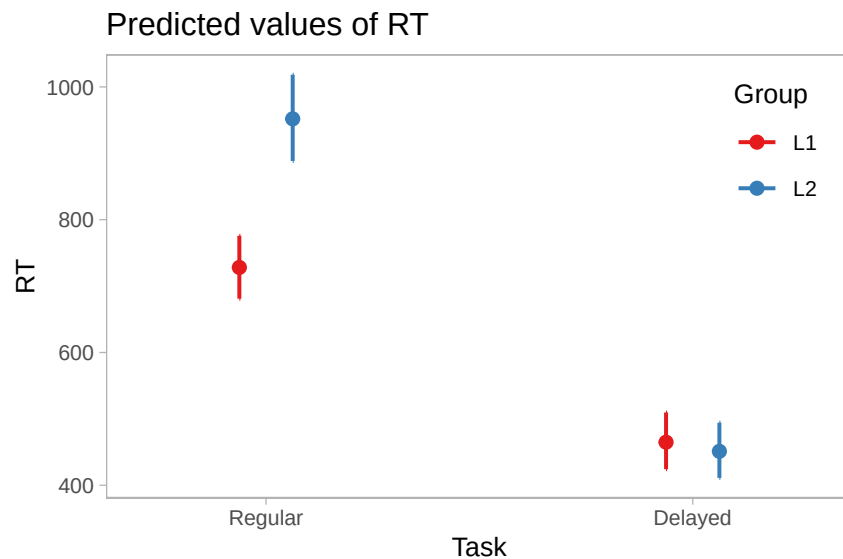


Figure 8. Predicted mean RTs for each group and naming task (lognormal model, stronger priors).

The model's predictions suggest that there is a large difference between the L1 and L2 groups in the regular naming task, but not in the delayed task. However, to conduct statistical inference, we must inspect the model's posterior distributions (which exist in the log-ms scale).

With the *ggdist* package, we can present several posterior distributions in the same plot, together with their means and 95% CrIs (Fig. 9):

```
ps.naming.stronger <- as_draws_df(m.naming.stronger)

pivot_longer(ps.naming.stronger,
  cols=starts_with("b_Task")) |> # Select parameters
  ggplot(aes(x=value, y=name)) + # Plot
  stat_halfeye(.point=mean)
```

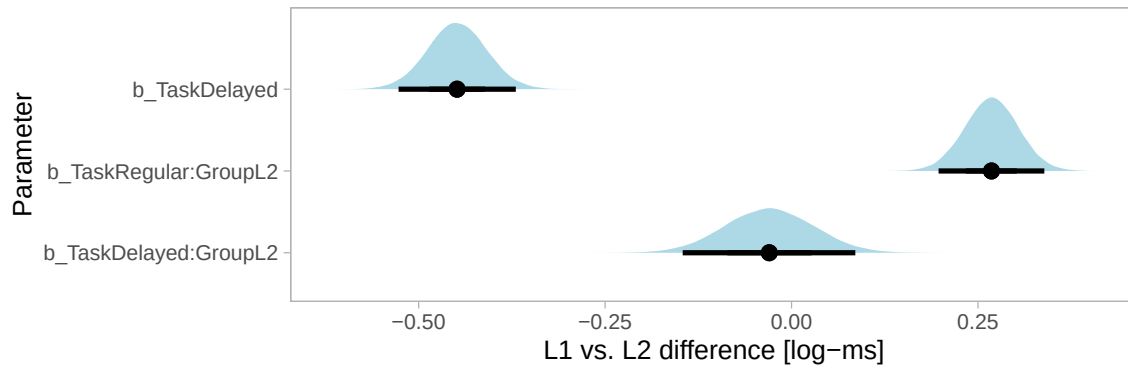


Figure 9. Posterior distributions of task and group effects (lognormal model, stronger priors).

The posteriors reveal a very large task difference in the L1 group, with delayed naming being much faster than regular naming. In addition, there is a very large (and relatively precise) L2 vs. L1 difference in the regular naming task. In contrast, the group difference in the delayed task has a close-to-zero mean and is much more uncertain, with 95% CrI bounds that span both negative (i.e., L2 faster than L1) and positive values (i.e., L2 slower than L1).⁵

Next, we calculate Bayes factors to test the hypotheses of a non-zero group difference (against the hypothesis of equality between groups), in each of the tasks:

```
hyp.regular <- "TaskRegular:GroupL2 = 0"
hyp.delayed <- "TaskDelayed:GroupL2 = 0"
hypothesis(m.naming.stronger, c(hyp.regular, hyp.delayed))
```

The tests produce an $\ln\text{BF}_{10}$ of 36.06 in the regular task, indicating very strong evidence for an L2 vs. L1 difference, but an $\ln\text{BF}_{10}$ of -1.26 in the delayed task, supporting the null hypothesis of equal mean RTs in the two groups.

In Bayesian models, further comparisons of interest can be easily obtained without needing to refit the model, nor having to resort to other packages. Any two parameters of interest can be compared by simply subtracting their posterior samples, yielding a posterior distribution for their difference. For example, by subtracting the two group effects, we obtain an estimate of the task $\times$ group interaction:

```
(ps.naming.stronger$b_TaskRegular:GroupL2 - ps.naming.stronger$b_TaskDelayed:GroupL2) |>
posterior_summary()
```

```
##      Estimate Est.Error Q2.5 Q97.5
## [1,]    0.298    0.0553 0.19 0.406
```

As expected, the estimate of the interaction is very large, since it amounts to a comparison between a large group effect (in the regular task) and a near-zero difference

⁵ The posteriors for random-effect and residual SDs can be visualised in a similar way (see Appendix S4).

(in the delayed task). We additionally use `hypothesis()` to obtain a Bayes factor testing the hypothesis of a non-zero interaction:

```
hypothesis(m.naming.stronger, "TaskRegular:GroupL2 - TaskDelayed:GroupL2 = 0")
```

The test produces an $\ln BF_{10}$ of 35.76, indicating very strong evidence for a task $\times$ group interaction. From these results, we would conclude that group differences are present *only* in immediate picture naming, not in delayed naming, which suggests that the L2 disadvantage is located at the earlier stages of language production (see Broos et al., 2021, for discussion).

Conclusions

We have shown with a range of practical examples how Bayesian analyses can be beneficial to the bilingualism researcher. Specifically, Bayesian methods: (a) direct our thinking and practice to the consideration of effect magnitudes and the uncertainty around them (e.g., Kruschke & Liddell, 2018); (b) can make use of prior information to arrive at more plausible and generalisable inferences that are less affected by outliers (e.g., Gelman & Tuerlinckx, 2000); and (c) allow the assessment of null differences between conditions or groups (e.g., Rouder et al., 2009). Given that much research in our field relies on between-group comparisons and involves difficult-to-reach and variable populations, Bayesian inference stands as a particularly useful framework for the bilingualism researcher.

Finally, an advantage of the Bayesian approach is its flexibility. Although the examples in this paper have made use of relatively simple models of lexical decision and naming times, Bayesian analyses are flexible enough to fit many different kinds of models and response distributions (e.g., Nicenboim et al., 2024). Indeed, the same framework and tools can be applied to the large variety of research questions and data types that are commonly analysed in our field (further examples can be found in Ciaccio & Veríssimo, 2022; Lago et al., 2023; Veríssimo, 2021a).

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Appendix:
Supplementary Materials for
“A gentle introduction to Bayesian statistics,
with applications to bilingualism research”

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Appendix S1. Sampling in Bayesian models

This section describes how to customise *brms* sampling, describes the effective sample size diagnostic of the number of posterior samples, and shows how to assess convergence in Bayesian models.

Customising sampling in *brms*

In Bayesian models, posterior samples are collected by separate ‘chains’. For every parameter in the model, each chain separately explores the parameter space and produces a different posterior distribution. What we call the posterior distribution of a parameter (containing, e.g., 4,000 samples) is actually the combination of several posterior distributions (e.g., 4 chains of 1,000 samples each). At the beginning of sampling, the chains start off in a region of the parameter space that may be quite far from the solution. Because of this, the initial portion of samples—also called the ‘warm-up’ samples—are discarded from the posterior distributions. We can define the number of chains, number of warm-up samples, and number of total samples by assigning values to the `chains`, `warmup`, and `iter` arguments of the `brm()` function (if they are not specified, *brms* assumes the following default values):

- `chains` (number of chains): 4
- `warmup` (number of warm-up samples per chain): 1000
- `iter` (number of total samples per chain, incl. warm-up): 2000

For each parameter, the number of usable (i.e., post warm-up) posterior samples is equal to $\text{chains} \times (\text{iter} - \text{warmup})$ (4,000 under the default values). While these defaults are generally appropriate when fitting simpler models (and when estimating posterior means), it may be desirable to increase the number of posterior samples to obtain more stable and accurate results. This is especially the case for estimating the bounds of 95% CrIs, which are at the tail of the posterior distribution and thus may show some instability when the number of posterior samples is small. In addition, computing Bayes factors absolutely requires the number of posterior samples to be increased from its default values (e.g., to 30,000–50,000 samples, Schad et al., 2022; Vasishth et al., 2018) to obtain stable results.

Effective Sample Size

A useful diagnostic of whether a parameter has enough posterior samples is given by the *effective sample size* (ESS) metric, which can be thought of as the number of posterior samples that provide unique information. Posterior samples tend to exhibit positive autocorrelation, so even though one may have collected thousands of posterior samples, the ‘effective’ number, when taking into account their autocorrelation, may in some cases be an order of magnitude smaller.

We can distinguish between ‘bulk ESS’ and ‘tail ESS’, both of which are shown in a *brms* model summary. The bulk ESS concerns the body of the posterior distribution and so it is more relevant when calculating measures of central tendency (e.g., mean and median).

The tail ESS relates to the tails of the posterior distribution and so it is more important for estimating the bounds of 95% CrIs. At the very least, both bulk and tail ESS should be larger than 100 per chain (400 in total when running 4 chains). *brms* helpfully outputs a “low effective sample size” warning when an ESS value is below this minimum. However, even without a warning, larger values may be desirable. Mixed-effects models, in particular, can exhibit low ESS for some parameters due to correlations between the posterior samples of fixed and random effects. When that is the case, a possible solution is to increase the number of samples in each posterior by assigning a larger value to `iter` and refitting the model (see Appendix S2 below for how to double the number of posterior samples by combining two models).

In order to prevent low ESS values and obtain accurate Bayes factors, the mixed-effects models in the main manuscript were actually fitted with 40,000 post warm-up samples per parameter (even though this was not made explicit in the code):

```
m.naming.stronger <- brm(formula.naming,
                          data=data.naming,
                          family=lognormal(),
                          prior=priors.naming.stronger,
                          sample_prior="yes",
                          chains=4, iter=12000, warmup=2000)
```

Assessing model convergence

The chains of a Bayesian model start exploring the parameter space from different initial values. In a well-behaved model, the chains quickly converge to the same solution and yield similar posterior distributions. One visualisation that is helpful for assessing convergence is the ‘traceplot’. This is a line plot of (post warm-up) parameter values over the order in which the samples were collected, overlaid for each chain. We can obtain traceplots for every model parameter by simply plotting the *brms* model object (e.g., with `plot(m.naming.stronger)`). Targeted traceplots can be obtained with the `mcmc_plot()` function of the *bayesplot* package as such (Fig. S1):

```
mcmc_plot(m.naming.stronger, type="trace",
          variable="^b_", regex=T,
          facet_args=list(ncol=1))
```

When all chains converge to a similar solution, the traceplots should give the impression of a ‘hairy caterpillar’. The traceplots for the fixed effects in the mixed-effects model reported in the main paper (lognormal model with stronger priors) are examples of the desired appearance (see Fig. S1).

We can also examine the posterior distributions of the fixed effects separately for each chain (Figure S2). This shows that all four chains produce posteriors that very closely overlap:

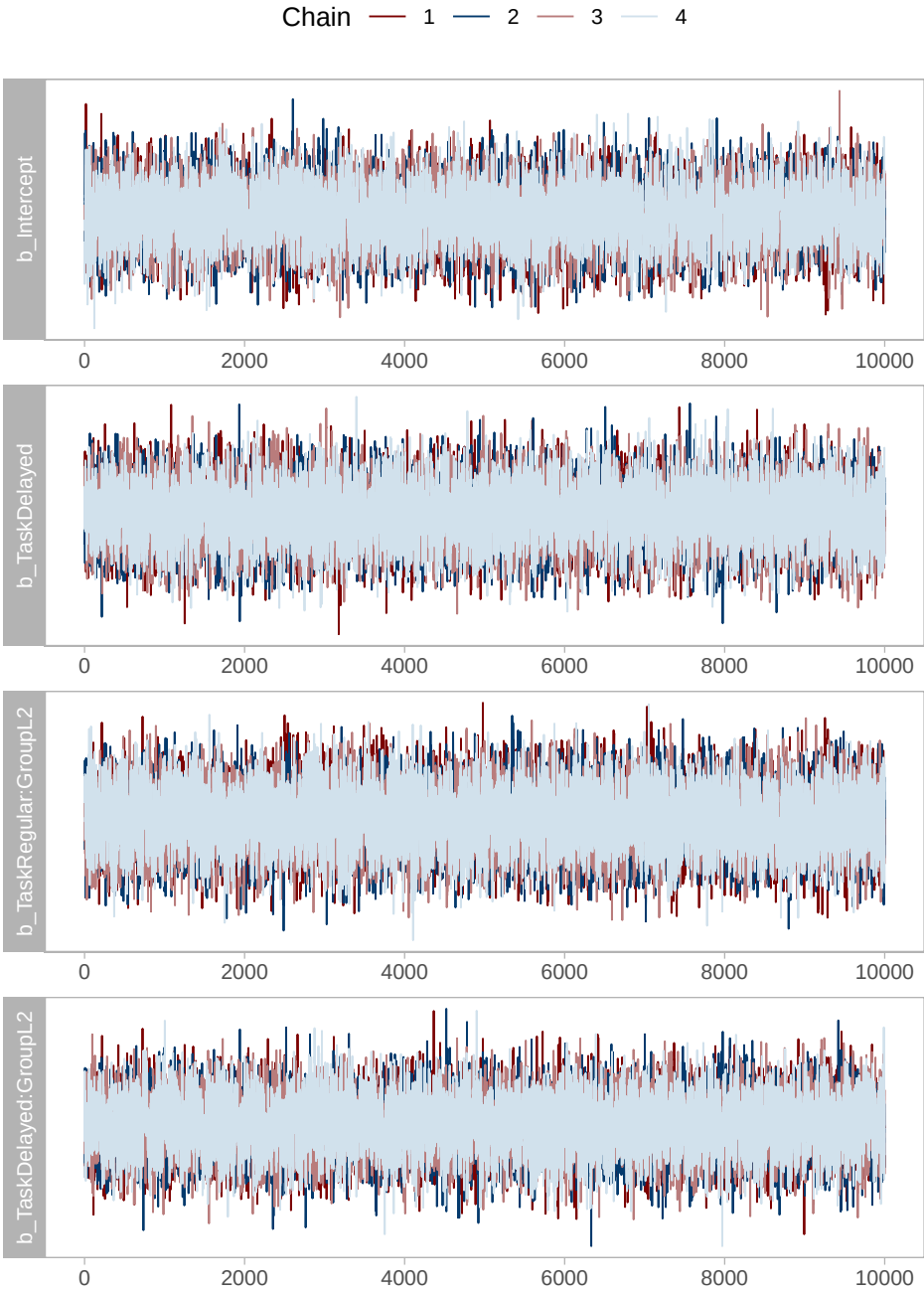


Figure S1. Traceplots of fixed effects for the lognormal model with stronger priors.

```
ps.naming.stronger <- as_draws_df(m.naming.stronger)           # Get posterior samples

pivot_longer(ps.naming.stronger, cols=c(starts_with("b_"))) |> # Select parameters
ggplot(aes(value, colour=as.factor(.chain))) +                 # Plot
geom_density(lwd=1) +
facet_wrap(vars(name), scales="free")
```

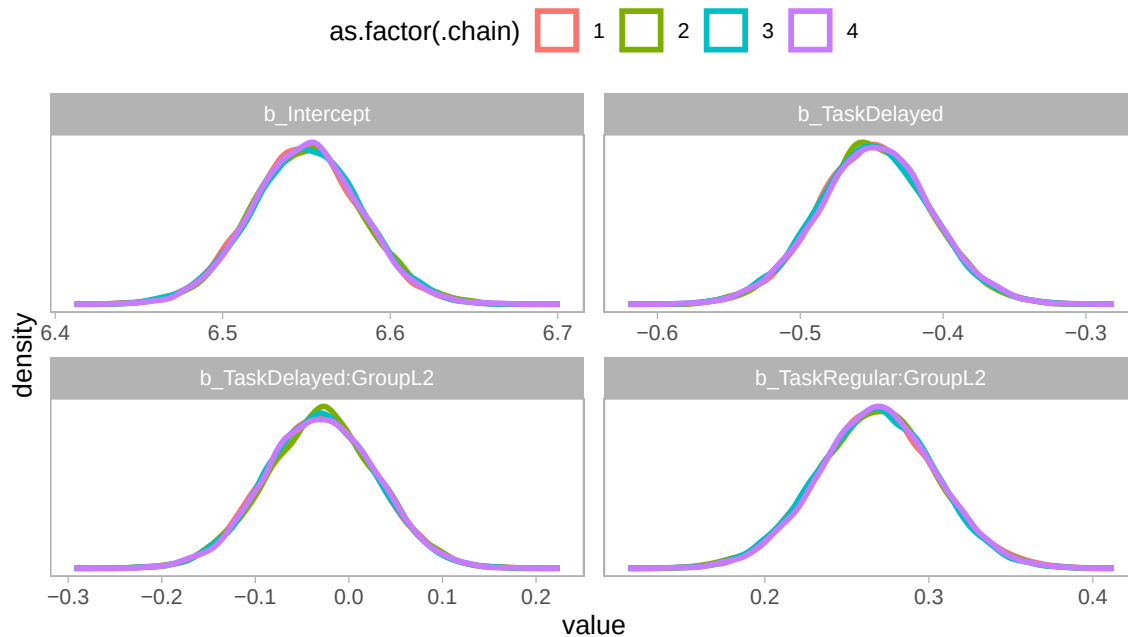


Figure S2. Posterior distributions of the fixed effects in the lognormal model with stronger priors, separately for each chain.

Besides visualisations, an important chain convergence diagnostic is the $\hat{R}$ statistic, which is based on a comparison of between- and within-chain estimates. $\hat{R}$ values should be between 1.00 and 1.01; if they are larger than 1.01, this indicates that the chains have not converged to the same solution. Note that the $\hat{R}$ statistic is not a formal check, nor a hypothesis test; it is only a quick diagnostic that can detect some convergence problems. Thus, even if the $\hat{R}$ of every parameter is very close to 1.00, this does not constitute a proof that the algorithm has converged, nor can it guarantee that there are no other problematic issues.

In models with many parameters, it may be useful to quickly isolate the most problematic cases. The `quickcheck()` function (available at <https://osf.io/n3jgm/>) can be applied to a *brms* model and returns the (two) parameters with the largest $\hat{R}$, the smallest bulk ESS, and the smallest tail ESS:

```
quickcheck(m.naming.stronger)
```

```
## $`Max Rhat`  
##                               Rhat  
## TaskDelayed                  1  
## Item.sd(TaskDelayed:GroupL2)  1  
##  
## $`Min Bulk ESS`  
##                               Bulk_ESS  
## Participant.cor(Intercept,TaskDelayed)  7424  
## Item.sd(TaskDelayed)                    9931  
##  
## $`Min Tail ESS`  
##                               Tail_ESS  
## Item.sd(TaskDelayed)                  8712  
## Item.sd(TaskDelayed:GroupL2)         10121
```

In this case, some of the random-effect parameters are somewhat more difficult to estimate, but all $\hat{R}$ and ESS values are perfectly appropriate.

Appendix S2. Computational efficiency

This section includes various techniques for increasing the speed and efficiency of Bayesian model fitting, including using the *cmdstanr* backend, parallelising model sampling over multiple CPU cores, saving models in order to safeguard a fit in the event of a crash, combining two identical models to double the number of posterior samples, reducing memory requirements, and fitting models as background jobs.

Backends

The *brms* package creates models in the Stan programming language, which actually does the model fitting for us. In order to communicate with Stan, *brms* needs a ‘backend engine’, which is *rstan* by default. An alternative backend is *cmdstanr*, which is generally faster and uses less memory. In order to use *cmdstanr*, we first need to install it. This can be achieved by the following commands:¹

```
install.packages("cmdstanr", repos = c("https://mc-stan.org/r-packages/",
                                       getOption("repos")))
library(cmdstanr)
check_cmdstan_toolchain(fix = TRUE)
install_cmdstan()
```

Once it is installed, we can use *cmdstanr* by passing `backend="cmdstanr"` to the `brm()` function. See <https://mc-stan.org/cmdstanr/articles/cmdstanr.html> for further documentation on installing and using *cmdstanr* as a backend.

Parallel sampling

Sampling can be sped up considerably by having chains run in parallel, rather than serially. We can detect the number of available cores in a system with:

```
(ncores <- parallel::detectCores())

## [1] 4
```

We can then assign each chain to its own CPU core by telling *brms* how many cores we want to use, as such:

```
m.naming.stronger <- brm(formula.naming,
                          data=data.naming,
                          family=lognormal(),
                          prior=priors.naming.stronger,
                          chains=4, iter=12000, warmup=2000,
                          cores=ncores)
```

It is even possible for a single chain to run in multiple cores, which can produce large speed-ups in systems with many CPU cores (see <https://cran.r-project.org/web/packages/b>

¹ On Windows, installing *cmdstanr* may require installing or updating *RTools*, which is available at <https://cran.r-project.org/bin/windows/Rtools/>.

rms/vignettes/brms_threading.html for a detailed description). As an example, if we wanted to run 4 chains on 8 CPU cores (i.e., each chain making use of 2 cores), we would run the following `brm()` call (note that within-chain parallelisation requires the *cmdstanr* backend):

```
m.naming.stronger <- brm(formula=naming,
                          data=data.naming,
                          family=lognormal(),
                          prior=priors.naming.stronger,
                          chains=4, iter=12000, warmup=2000,
                          cores=4, threads=threading(2),
                          backend = "cmdstanr")      # CPU cores = cores * threading
```

Saving models to a file

Fitted models that exist only as objects in the R environment are vulnerable to R session crashes or other computer issues. Since fitting Bayesian models can take a long time, it is safer to save them as soon as they have finished fitting. A *brms* model can be saved to an `.rds` file after fitting by specifying a filename in the `file` argument to `brm()` (note that the `.rds` extension is added automatically):

```
m.naming.stronger <- brm(formula=naming,
                          data=data.naming,
                          family=lognormal(),
                          prior=priors.naming.stronger,
                          chains=4, iter=12000, warmup=2000,
                          file="models/m_naming_stronger")
```

When the `brm()` command above is run and the file does not yet exist at that location, the model is fitted and immediately saved. If the file already exists, the fitted model is loaded from the `.rds` file instead of refitted. This behaviour can be controlled by the `brms.file_refit` option. For example, setting `options(brms.file_refit="on_change")` fits the model whenever the file does not exist *or* if the model (or the data) have changed relatively to the saved model.

Combining two models

It is possible to increase the number of posterior samples in a model by fitting a new model and combining it with the previous one. This is particularly useful when a model displays low ESS for some parameters (see above). We do this using `combine_models()`:

```
model.combined <- combine_models(model1, model2)
```

For the model combination to work, the two models need to be identical in their formula, parameters, and even in the number of posterior samples. Thus, one can use `combine_models()` to effectively double the number of posterior samples in a model, while avoiding having to perform a full refit from scratch.

Reducing memory requirements

One of the main difficulties when fitting Bayesian models is that they tend to have heavy (RAM) memory requirements, especially in the case of mixed-effects models with many parameters or when analysing large datasets. One technique to reduce memory requirements is to make use of ‘sparse’ matrices for the fixed-effect predictors. This can be defined in the model formula as such:

```
# Formula
formula.naming <- bf(RT ~ Task/Group +
  (1+Task|Participant) +
  (1+Task/Group|Item),
  center=FALSE,
  sparse=TRUE)
```

This option will reduce the model’s RAM memory requirements when the fixed-effects contrast matrix contains many zeros. The contrast matrix for the mixed-effects model in the main paper can be obtained in the following way:

```
# Formula
mm <- make_standata(formula(m.naming.stronger), m.naming.stronger$data)
unique(mm$X)
```

##	Intercept	TaskDelayed	TaskRegular:GroupL2	TaskDelayed:GroupL2
## 2492	1	0	1	0
## 2491	1	1	0	1
## 2577	1	0	0	0
## 1748	1	1	0	0

Indeed, for these contrasts, there is a large number of zeros in the predictor values, so we can expect the use of sparse matrices to reduce memory requirements. However, note that using sparse matrices will not speed up model fitting (and it might even make it slower).

A second technique for reducing the memory requirements of Bayesian mixed-effects models is to tell *brms* not to save each random effect. By default, a *brms* model contains full posteriors for each random-effect level (e.g., for each participant and item). If these are not needed, we can avoid storing every random effect inside the model object using the `save_pars` argument of `brm()`:

```
m.naming.stronger <- brm(formula.naming,
  data=data.naming,
  family=lognormal(),
  prior=priors.naming.stronger,
  chains=4, iter=12000, warmup=2000,
  save_pars = save_pars(group = FALSE))
```

A model fitted in this way will still contain a full posterior for all variance components (i.e., for the random-effect SDs and correlations), but not for each individual participant and item adjustment.

Fitting models as background jobs

It is often useful to have models fit in the background, while still being able to use the R console or running other scripts. One option that is provided by default in the *RStudio* interface is to source scripts as background jobs. For greater flexibility, we recommend the *job* package (Lindeløv, 2021). The simplest way to use it is to wrap the `brm()` function inside `job()`. The model will be fitted in the background and the model object will be sent to the environment when it has finished fitting (here, 2 CPU cores are left unused, so that they remain free for other tasks):

```
job({  
  ncores <- parallel::detectCores()  
  m.naming.stronger <- brm(formula.naming,  
                           data=data.naming,  
                           family=lognormal(),  
                           prior=priors.naming.stronger,  
                           sample_prior="yes",  
                           chains=4, iter=12000, warmup=2000,  
                           cores=ncores-2,  
                           file="models/m_naming_stronger")  
})
```


Appendix S3. Visualisation of LKJ priors

The *LKJ* (Lewandowski-Kurowicka-Joe) distribution is defined over symmetric matrices with unit diagonals, such as the matrices of random-effect correlations between. The probability density of *LKJ* distributions peaks at zero and the distribution's 'shape' governs how steeply the density declines into the larger correlation magnitudes. Figure S3 shows three different *LKJ* distributions for a 2x2 correlation matrix with shapes 1, 2, and 3 (i.e., with 2 random effects). Note that the particular prior distribution that is assigned to the correlation values additionally depends on the size of the matrix, that is, on the number of correlations being estimated (larger matrices exhibit a narrower *LKJ* for the same shape parameter). In the mixed-effects models described in the main manuscript, an *LKJ* distribution with shape 2 was used as a weakly-informative prior on all random-effect correlations.

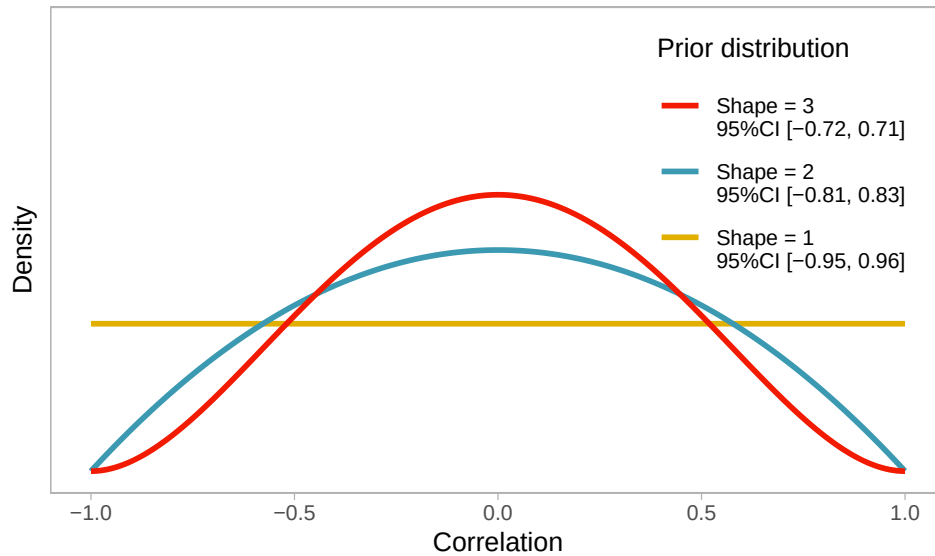


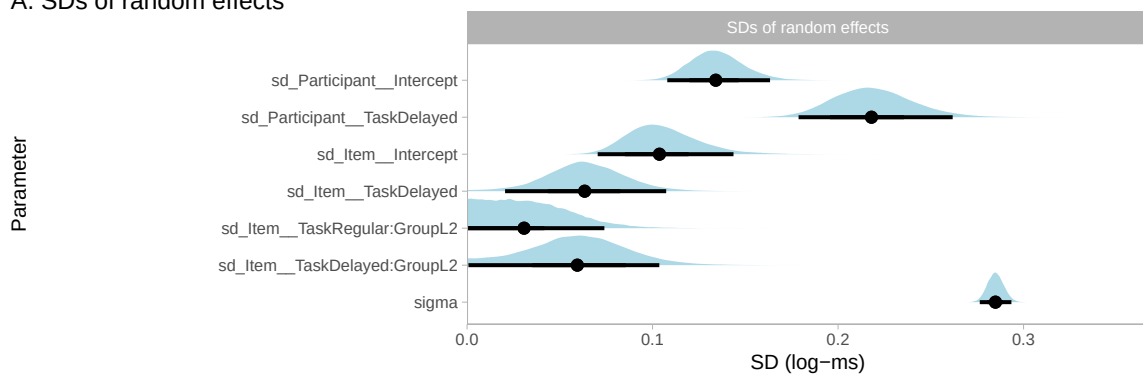
Figure S3. Examples of *LKJ* distributions with shapes 1, 2, and 3, for a 2x2 correlation matrix (plotted using the `dlkcorr_marginal()` function from the *ggdist* package).

Appendix S4. Posteriors for variance components

In the main manuscript, we have demonstrated how to visualise the posteriors of fixed effects using the *ggdist* package. We can obtain posteriors for the random effects and residual SDs in a similar way. However, since posteriors for random effects can be quite skewed, we instead ask for medians and highest-density intervals (in this case for the lognormal model with stronger priors that is described in the main manuscript):

```
pivot_longer(ps.naming.stronger,
             cols=c(starts_with("sd_"), "sigma")) |> # Use "cor_" for correlations
ggplot(aes(x=value, y=name)) +
stat_halfeye(.point=median, .interval=hdci)
```

A. SDs of random effects



B. Correlations between random effects

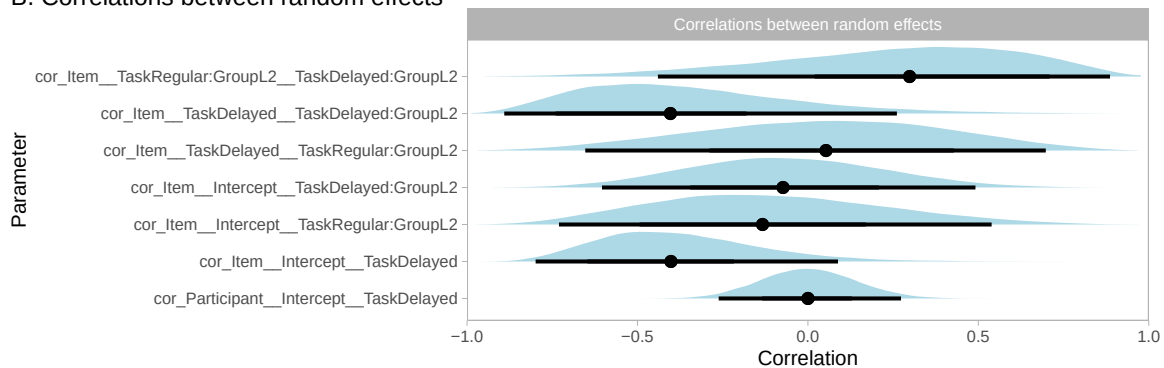


Figure S4. Posterior distributions for (A) random intercepts and slopes and residual SD, and (B) correlations between random effects.

Figure S4A displays the posteriors for the random and residual SDs; Figure S4B displays the posteriors for the random-effect correlations. The figure shows that the largest source of variance is trial-by-trial variation (σ), but there is also substantial between-participant variability in the effect of task. This variation indicates that some participants show very large differences between regular and delayed naming, while others show much smaller differences. This may ultimately be attributed to differences in participant variables—such as language proficiency—which could be further investigated.

It is also noticeable that the posteriors for the random-effect correlations (S4B) are all very wide (i.e., uncertain), spanning both negative and positive values. This is a common occurrence in Bayesian mixed-effects models, because random-effect correlations are hard to estimate appropriately—at least with the sample sizes that are commonly used in psychology and linguistics. However, it is reassuring that the Bayesian model is able to appropriately estimate this uncertainty. In contrast, frequentist mixed-effects models with complex random-effect structures often yield “singular fits”, which indicate difficulties with the estimation of random parameters that are very uncertain or close to zero (e.g., Bates et al., 2018).

References

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